

Non-anthropogenic metric value = 40.

Anthropogenic target metric value = 45.

Percentage reduction required = $((70-45)/70) = 35.7\%$

All of the hourly anthropogenic observations are reduced by 35.7%. The average of the first 8 values (the window on which the Test metric is based) will be exactly 45, the anthropogenic target metric value. Finally, the adjusted hourly observations are calculated by adding the non- anthropogenic hourly observation to the adjusted hourly anthropogenic observations.

A.3.2.1.2 Example: Some Hourly Observations are Below the Intraday Background (Multiple Iterations Required)

In the above example, the anthropogenic target metric value was met on a single iteration because all of the hourly observations were greater than the Intraday Background Level. In this case, a simple percent reduction of all hourly values will produce an average in the window that is equal to the anthropogenic target metric value. If some of the hourly observations in a day are less than or equal to the Intraday Background Level, however, then BenMAP uses an iterative procedure.

On each iteration, it adjusts hourly observations using the 10-step method given above. It then compares the new metric value to the target metric value. If the difference is less than or equal to 0.05 ppb, the rollback procedure is finished. Otherwise, another iteration is required. The iterative procedure is illustrated in the following example.

Suppose that:

Metric = EightHourDailyMax,

Target metric value for a given day = 85

Intraday Background Level = 40.

Suppose also that the hourly observations on that day are:

530	20	25	60	35	35	40	60	70	100	100	100	100
100	100	100	100	60	33	40	30	30	25	20		

Then, we know that the 8-hour daily maximum = 100.6.

Non-Anthropogenic Hourly Observations, Iteration One:

40	20	25	40	35	35	40	40	40	40	40	40	40
40	40	40	40	40	40	33	40	30	30	25	20	

Anthropogenic Hourly Observations, Iteration One:

490	0	0	20	0	0	0	20	30	60	60	60	60	60
60	60	60	20	0	0	0	0	0	0				

Non-Anthropogenic Metric Value: 34.4 (EightHourDailyMax - calculated over the same eight hour window as the initial metric value was calculated over)

Anthropogenic Metric Value: 66.3

Anthropogenic Target Metric Value: 50.6

Percentage Reduction Required: 23.6%

Reduced Anthropogenic Hourly Observations, Iteration One:

374	0	0	15	0	0	0	15	23	46	46	46		
46	46	46	46	46	15	0	0	0	0	0	0	0	

Reduced Hourly Observations, Iteration One:

414	20	25	55	35	35	40	55	63	86	86	86	86	
86	86	86	86	55	33	40	30	30	25	20			

Reduced Metric Value (EightHourDailyMax): 85.8

Target Metric Value (EightHourDailyMax): 85

Non-Anthropogenic Hourly Observations, Iteration Two:

40	20	25	40	35	35	40	40	40	40	40	40	40	
40	40	40	40	40	33	40	30	30	25	20			

Anthropogenic Hourly Observations, Iteration Two:

374	0	0	15	0	0	0	15	23	46	46	46	46	
46	46	46	46	15	0	0	0	0	0	0	0		

Non-Anthropogenic Metric Value: 40 (EightHourDailyMax - calculated over the same eight hour window the initial metric value was calculated over)

Anthropogenic Metric Value: 45.8

Anthropogenic Target Metric Value: 45

Percentage Reduction Required: 1.9%

Reduced Anthropogenic Hourly Observations, Iteration Two:

368	0	0	15	0	0	0	15	23	45	45	45	45	45
45	45	45	15	0	0	0	0	0	0				

Reduced Hourly Observations, Iteration Two:

408	20	25	55	35	35	40	55	63	85	85	85	85	85
85	85	85	55	33	40	30	30	25	20				

Reduced Metric Value (EightHourDailyMax): 85

The above example, in addition to illustrating the Intraday Percentage Rollback, also illustrates one reason why the iterative procedure can be necessary. When using the EightHourDailyMax metric in the Attainment Test, it is possible for the window over which the maximum eight hour average occurs to move after a single pass through the rollback procedure. When this happens, it becomes necessary to go through additional iterations to hit the target metric value.

A.3.3 Intraday Rollback - Incremental

To adjust hourly observations using Incremental rollback, BenMAP calculates the increment required to reduce the anthropogenic metric value to exactly the anthropogenic target metric value. This incremental reduction is then applied to all of the anthropogenic observations (but - they are not allowed to fall below zero). Finally, these reduced anthropogenic observations are added to the non-anthropogenic observations to give the final reduced observations.

Example:

Initial Hourly Observations:

20	20	25	60	35	35	40	70	35	30	65	90	76
65	35	35	54	60	33	40	30	30	25	20		

Initial Metric Value (EightHourDailyMax): 60

Target Metric Value (EightHourDailyMax): 55

Intraday Background Level: 40

Intraday Rollback Method: Incremental

Non-Anthropogenic Hourly Observations, Iteration One:

20	20	25	40	35	35	40	40	35	30	40	40	40
40	35	35	40	40	33	40	30	30	25	20		

Anthropogenic Hourly Observations, Iteration One:

0	0	0	20	0	0	0	30	0	0	25	50	36
25	0	0	14	20	0	0	0	0	0	0		

Non-Anthropogenic Metric Value (EightHourDailyMax): 38.8

Anthropogenic Metric Value (EightHourDailyMax): 21.3

Anthropogenic Target Metric Value (EightHourDailyMax): 16.3

Incremental Reduction Required: 5.0

Reduced Anthropogenic Hourly Observations, Iteration One:

0	0	0	15	0	0	0	25	0	0	20	45	31
20	0	0	9	15	0	0	0	0	0	0		

Reduced Hourly Observations, Iteration One:

20	20	25	55	35	35	40	65	35	30	60	85	71
60	35	35	49	55	33	40	30	30	25	20		

Reduced Metric Value (EightHourDailyMax): 56.25

Target Metric Value (EightHourDailyMax): 55

Non-Anthropogenic Hourly Observations, Iteration Two:

20	20	25	40	35	35	40	40	35	30	40	40	40
40	35	35	40	40	33	40	30	30	25	20		

Anthropogenic Hourly Observations, Iteration Two: